

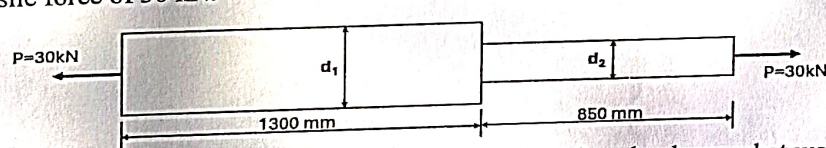


Shri Guru Gobind Singh Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Production Engineering
End Semester Examination
Academic Year: 2025-26 (semester I)

Class: SY B.Tech.	Course: Strength of Materials	Course Code: PCC-PE201
Date: 26.11.2025	Time: 01.30 pm to 03.30 pm	Max. Marks: 60

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Solve any **SIX** questions.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.

Q.N.		Marks	CO	BT
Q.1	In a torsion test, a shaft of diameter 25 mm is twisted 0.025 radians over a length of 200 mm when a torque of 0.4 kNm was applied. Evaluate the Modulus of rigidity of the material.	10	CO3	BT3
Q.2	A solid circular shaft transmits 70000 W power at 145 rpm. Calculate the shaft diameter if the twist in the shaft does not exceed 1.5 degrees in 1.85 meters length of the shaft, and shear stress is limited to 48 MN/m ² . Take C= 110 GN/m ² .	10	CO4	BT3
Q.3	A girder of uniform cross section and constant depth is freely supported for 3 meters. If the point load at the midpoint is 30 kN and MI is $15.614 \times 10^{-6} \text{ m}^4$, calculate the central deflection and the slopes at the ends of the beam. Take E= 200 GN/m ² .	10	CO4	BT3
Q.4	A steel cantilever with a span of 1.5 m carries a point load of W kN at its free end. The moment of inertia of the section of the cantilever is 8900 cm ⁴ . If the deflection at the free end does not exceed 0.75 cm, what must be the value of W? Take E =210 GN/m ² .	10	CO5	BT3
Q.5	A rod is 1.85 m long at 9.5°C. Find the expansion of the rod when the temperature is raised to 80°C. If this expansion is prevented, find the stress in the material. Take E = $1 \times 10^5 \text{ N/mm}^2$. and coefficient of thermal expansion = 0.000012 per °C.	10	CO3	BT3
Q.6	A steel plate is bent into a circular arc of radius 10 meters. If the plate section is 90 mm wide and 10 mm thick, find the maximum stress induced and the bending moment which can produce this stress. Take E= $2 \times 10^5 \text{ N/mm}^2$.	10	CO4	BT3
Q.7	A steel bar 2150 mm long is 20 mm in diameter for half the length and 10 mm diameter for other half length. Find the extension of the bar when subjected to a tensile force of 30 kN.	10	CO2	BT3
 For a steel bar of uniform diameter having same length and volume what would be the extension when subjected to the same tensile force? Take E = $2 \times 10^5 \text{ N/mm}^2$				
Q.8	A point is subjected to both tensile perpendicular stresses of 50 MN/m ² and 30 MN/m ² . Calculate the normal stress and the shear stress on a plane inclined at 30° to the axis of the second stress. Use analytical and Mohr's circle method.	10	CO1	BT3

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

On successful completion of this course, students should be able to:

- CO 1 To differentiate the mechanical behaviour of engineering materials.
- CO 2 To apply the theories of elastic failure.
- CO 3 To determine the shear force and bending moment distribution of the beam and locate the point of contraflexure.
- CO 4 To calculate stresses in beams for various boundary conditions.
- CO 5 To determine the slope and deflection of the beam for various boundary conditions.